

- 4 Ribulose 1,5-bisphosphate carboxylase/oxygenase (rubisco) is an important enzyme involved in the light independent stage (Calvin cycle) of photosynthesis. It fixes carbon by combining carbon dioxide with RuBP.

In certain situations, the active site of rubisco becomes occupied by a sugar phosphate, making the enzyme inactive. Rubisco can become active again in the presence of another enzyme, rubisco activase.

- (a) Name all the bonds that are likely to hold a molecule of rubisco in shape.

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- (b) Suggest how rubisco activase can activate rubisco.

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- (c) C4 plants such as maize have adaptations that allow them to have high rates of carbon fixation at high temperatures. Without these adaptations, some plants (C3 plants) are affected at high temperatures by a process known as photorespiration.

In photorespiration, rubisco combines oxygen with RuBP. This leads to a decrease in the rate of photosynthesis.

- (i) Describe **and** explain how the anatomy of the leaves of C4 plants such as maize allows them to have high rates of carbon fixation at high temperatures.

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- (ii) In C3 plants, the rate of photorespiration increases at high light intensities as well as at high temperatures.

Suggest why the rate of photorespiration increases at high light intensities in C3 plants.

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- (iii) Explain why the rate of photosynthesis decreases as a result of photorespiration in C3 plants.

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